

Build. Code. Deliver.

Teacher guide • setup, facilitation, answer key and rubric

Grades 6-8 • 8 modules • 16 sessions of 45 minutes

LEGO Education Computer Science & AI Kit 6–8 (#45522, 2026)



Imaginative LEGO-style mission concept. Follow the official #45522 build resources.

Predict. Build. Test. Explain. Improve.

Design a brick-built delivery rover. Carry a paper parcel, recognize a colored delivery marker, and explain the evidence behind each improvement.

Set up for a dependable lesson

Scope and readiness

This is a teaching draft for teams of four in grades 6-8. It assumes basic computer navigation, but no previous robotics coding. Reading and multiplication support may be needed. Complete the teacher hardware preflight before classroom use; these block recipes have not been tested on a physical robot in this project.

Materials and preparation

For each team of four: one #45522 kit and one compatible device with LEGO Education Coding Canvas. Rotate builder, programmer, test operator and evidence recorder so each learner practices every role.

Shared supplies: ruler, masking tape, matte white and red paper, light cardboard, scissors and removable adhesive. Mark a flat floor lane with a wide red finish strip and a generous empty run-out area. No stairs, table edges or people as targets.

Before class, use LEGO's #45522 building instructions and Teacher Portal to assemble and check a double-motor wheeled base. Secure a downward-facing color sensor ahead of the drive wheels. Photograph the actual classroom build and record its sensor height; concept artwork is not a parts-by-parts assembly guide. If the school has no checked base, schedule a separate teacher build/preflight session.

Open code.legoeducation.com. Follow LEGO's getting-started flow to connect the team's double motor and color sensor using its connection card. Verify that only the intended team hardware responds. Keep the program running on the connected device during trials. Use the downloadable lesson recipe to assemble word blocks; recipes describe logic rather than claiming to be importable LEGO project files.

Charge hardware using LEGO's supplied charging instructions before class. Check the program stop control and the manufacturer's hardware power-off procedure with wheels raised. Use the teacher-approved base, save the project locally, and test forward direction at low speed before floor runs.

Run the classroom

Before learners arrive

For every team: check the actual build, charge, hardware pairing, saved block project, forward direction, on-screen stop, power-off, stationary red/white/unknown readings, and a low-speed marker-stop trial. Mark separate floor lanes. Reserve one known-working robot for diagnosis. Label each kit and keep its calibration log with it.

Safety and supervision

Every powered trial needs teacher approval, a clear floor lane, low speed and an operator beside the stop control. Stop the program immediately on unexpected motion; use the hardware power-off procedure if software or the connection fails. The supplied recipes require a short maximum trial duration. Verify these controls on the real kit before teaching.

Keep hair and fingers clear of moving parts. Power down before changing the build. Stop on unrecognized color, disconnection or loss of control. A color marker is a delivery cue, not an obstacle detector or safety system. Follow school and manufacturer procedures.

Session rhythm

Session A: launch and predict (5 min), explain/model (10), build and program (20), first test (5), save and tidy (5).

Session B: retrieval (5 min), fair tests and revisions (25), explanation or peer review (10), exit ticket and tidy (5).

Support and extension

Provide printed arrow cards and teacher-prettested block projects. Read prompts aloud or use narration and transcripts. Permit diagrams plus an oral explanation instead of extended writing. Rotate roles so every learner programs and explains. For advanced learners, require a new constraint and an evidence-based test, rather than simply increasing speed.

If hardware is unavailable

Use the grid simulator for sequences, decisions and evidence logs, plus paper cradle prototypes. Label these as simulated or paper results. Schedule the physical build, measurement and stop checks when kits are available; simulator completion does not establish physical robotics proficiency.

Assessment evidence

Collect an individual exit explanation each module, a team trial log, and the final handover. Grade engineering reasoning and honest evidence as well as whether the rover completes a run.

Course map

Mission	Physical deliverable	Concept
1. Wake up your rover	Build a stationary input-to-output response	Input / event / output
2. Motor checkout	Make a controlled, repeatable drive	Drive command and measurement
3. Design a cargo cradle	Carry a paper parcel without blocking the robot	Constraints and attachment design
4. Crack the color code	Build and test a dependable marker reader	Sensor calibration and uncertainty
5. Code a delivery route	Drive to a marked delivery bay	Sequence and open-loop control
6. Nail the delivery stop	Stop on a colored finish marker	Conditional and latched stop
7. Plan the delivery mission	Deliver a paper parcel using a tested approach	Integration and acceptance tests
8. Showcase with evidence	Prove, explain and hand over your robot	Handover and evidence

Scheduling options

Standard: two sessions per week for eight weeks. Intensive: four sessions per week for four weeks. Keep the repeated trials and reflections in either schedule. Allow an extra setup session if the kits arrive unassembled.

Wake up your rover

Learning evidence

Identify input, rule and output, and demonstrate three predictable color-to-screen responses without wheel motion.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Three correctly explained input-to-screen tests with motors stationary.

Exit question and answer key

A remotely operated exploration robot is still a robot. What extra capability would make this rover autonomous?

It would read an environmental sensor and use a programmed rule to choose an action without a person choosing every movement.

Misconception to address

The sensor detects reflected light; it does not understand the meaning of a delivery.

Troubleshooting

Check the paired team hardware and sensor view. If wheels move, stop, remove movement blocks and ask the teacher to inspect.

Stretch task

Design a message that also uses words or symbols so a learner need not distinguish screen colors.

Motor checkout

Learning evidence

Run a bounded forward drive, demonstrate stop, and identify two sources of variation.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Three bounded drives with automatic stops, plus an operator-stop test and distance records.

Exit question and answer key

Why can the same motor command produce a different distance tomorrow?

Surface, charge, alignment and payload affect movement. The motor setting is not a measured travel speed.

Misconception to address

Equal speed commands do not guarantee perfectly equal wheel motion.

Troubleshooting

Check charge, connection card, selected motor and wheel clearance. Recheck forward direction with wheels raised.

Stretch task

Compare two low motor commands using three trials each and mean distance.

Design a cargo cradle

Learning evidence

Build and compare two removable cargo cradles using consistent trials and a stated design constraint.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Parcel stays in place for three short trials; the sensor view, wheels, controls and switch remain unobstructed.

Exit question and answer key

Why should you keep the parcel and code unchanged when comparing the cradles?

It helps isolate the effect of the cradle design; changing several factors would make the cause of a difference unclear.

Misconception to address

The supplied chassis is already built. Bracing a cardboard attachment is different from redesigning the robot’s electronics.

Troubleshooting

If the attachment shifts or interferes, stop and remove it. Reduce size or reposition it; do not compensate by increasing speed.

Stretch task

Have another team reproduce your attachment from the drawing, then compare results.

Crack the color code

Learning evidence

Compare repeated color classifications and select a marker setup using recorded evidence.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Nine initial readings, a controlled comparison, and a justified marker choice with a limitation.

Exit question and answer key

The same red card is reported correctly only twice in three trials. What should you do before driving?

Investigate height, light and material with motors stopped, change one factor and retest. Two correct readings do not establish dependable recognition.

Misconception to address

A color sensor does not measure the distance to a wall. Unknown is not evidence that the path is clear.

Troubleshooting

Secure the sensor mount, remove glare, keep the sample flat and verify pairing. Compare another matte sample before changing several variables.

Stretch task

Compute the proportion of correctly classified samples and discuss why three trials are limited evidence.

Code a delivery route

Learning evidence

Program a two-leg route and use three trials to tune one timing variable.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

A documented two-leg route ends within the team’s stated bay tolerance in at least two of three trials, and stops every time.

Exit question and answer key

Why can a route succeed once but miss the bay on the next run?

Open-loop timing cannot correct for differences in traction, batteries, start direction or load; repeated trials reveal reliability.

Misconception to address

A “turn right” instruction in a grid represents a precise quarter-turn; a real motor needs calibration or feedback.

Troubleshooting

Shorten the sequence and test its legs separately. Check the start direction before changing the code.

Stretch task

Compare a long route with a shorter route. Keep a record of program length and successful trials.

Nail the delivery stop

Learning evidence

Use a repeated color decision to stop on a red marker in three supervised trials and measure stopping position.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

Three marker-triggered stops with measured positions, no automatic restart and no motion on an unrecognized reading.

Exit question and answer key

Why can a narrow red strip be missed even when the stationary test worked?

The robot moves between readings. It may cross a narrow strip before the next reading, or lighting and height may change. Lower speed and a wider marker create more opportunity to detect it; real testing is still required.

Misconception to address

Recognizing a floor marker does not detect obstacles, people or table edges.

Troubleshooting

Return to stationary color tests. Check the mount, marker width and program order. A stopped trial must not restart simply because white becomes visible again.

Stretch task

Add distinct messages for delivery marker, unknown reading and time limit, without restarting movement.

Plan the delivery mission

Learning evidence

Combine a cargo attachment, a bounded route and a sensor stop into a mission with measurable acceptance criteria.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

At least two of three full mission trials meet all three criteria; every trial remains bounded and supervised.

Exit question and answer key

Your first version passed 1 of 3 trials and your revision passed 2 of 3. What can you claim?

The revision performed better in these small trial sets. More trials are needed before claiming broad reliability; do not claim it always works.

Misconception to address

A visually impressive run is not enough; the criteria must be set before testing.

Troubleshooting

Test cargo, movement and color recognition separately. Recheck the actual lane lighting before combining them.

Stretch task

Design a sorting mission using two colored bays. Explain what added logic and testing would be needed.

Showcase with evidence

Learning evidence

Present a reproducible demonstration and defend a design choice using measurements and a limitation.

Session A: model, then build

Use the build card in the student workbook. Ask for a prediction before showing the result. Model the first step and name the input, decision or physical constraint. Teams complete the build and one initial test; inspect stop access and setup before powered trials.

Session B: test, revise, explain

Teams collect the required repeated trials, revise one factor, and compare. Circulate with “What stayed the same?” and “What evidence supports that change?” Close with the question below and save the program version.

Look for

A reproducible handover, three recorded trials, a two-minute evidence-based explanation and an individual reflection.

Exit question and answer key

What should you say if the showcase demonstration fails?

Describe what happened, stop safely, compare it with the prior trial record and identify a specific next test. Never hide or replace the recorded result.

Misconception to address

Engineering success includes finding and explaining limits, not only achieving a perfect demonstration.

Troubleshooting

Use the saved program and setup drawing to reproduce conditions. If time is short, demonstrate a verified subsystem and report the full mission’s limits honestly.

Stretch task

Propose one additional user requirement, explain its tradeoff and design a test without claiming the feature is already implemented.

Assess the work, not the spectacle

Score each dimension 1-4 (16 points total). Use individual explanations to distinguish understanding within a team. A teacher intervention for a safety issue prompts reteaching before another powered run.

Dimension	4: Independent	3: Secure	2: Developing	1: Beginning
Mission & constraints	Measurable criteria; reproducible safe setup.	Clear criteria; minor setup clarification.	Some criteria vague or omitted.	Criteria or operating plan absent.
Program explanation	Accurate sequence and conditional; explains limits.	Explains input, rule and output correctly.	Partial explanation; needs prompts.	Cannot connect code to behaviour yet.
Testing & iteration	Repeated fair trials; justified revision and limitation.	Records repeated trials; uses evidence.	Incomplete controls or records.	Claims without usable records.
Communication & roles	Peer can reproduce; each learner explains their work.	Clear handover; shared roles.	Handover or participation uneven.	Handover unclear; little individual evidence.

Sample feedback

“Your stop rule is clearly explained. Your three trials changed both speed and marker width, so the evidence does not identify the cause. Keep speed fixed and compare two marker widths with three trials each.”

Optional color-sorting investigation

With motors stopped, compare two colored parcel labels and an unfamiliar sample. Design a table mapping each reliable reading to an action. Explain how unknown samples are handled. This is an investigation extension; sorting hardware and programs need separate design and testing.

Mission Lab and portfolio handover

Simulator facilitation

Students choose a map, write a prediction, add commands, step through the result, record a trial and download the CSV log. Turns are exact quarter-turns; forward is exactly one cell. “If blocked” checks the adjacent cell, then turns right when blocked or moves forward when clear. It does not continuously drive around obstacles.

Teacher solution routes

First delivery: F F F R F F F F.

Around the corner: F R F F F R F F.

Sensor decision: F I F F F F.

Explain the portfolio honestly

This sample demonstrates curriculum design, alignment of practice and assessment, original AI-generated concept visuals, synthesized narration, and an interactive logic model. Present physical-robot calibration and classroom validation as next steps. This custom web course does not establish Storyline/Rise authoring proficiency by itself; the original course is a Rise export.

Suggested interview walkthrough (4 minutes)

1. Explain the learner and performance goal. 2. Show a build card and its matching evidence log. 3. Demonstrate a failed simulator route and the learner feedback. 4. Show the teacher rubric, narration/transcript and one visual; explain the hardware-testing boundary.

Sources and credits

Selected kit and current US pricing

<https://education.lego.com/en-us/products/lego-education-computer-science-and-ai/45522/>

Official #45522 build instructions

<https://www.lego.com/service/buildinginstructions/45522>

Coding Canvas

<https://code.legoeducation.com/>

Teacher Portal and LEGO lessons

<https://teach.legoeducation.com/>

LEGO Education Python reference (optional advanced route)

<https://github.com/LEGO/LEGOEducation>

SPIKE retirement and support

<https://education.lego.com/en-us/spike-update-2026/>

Original Rise course

<https://luvbniz.github.io/robotics/>

Hardware and US prices checked September 10, 2026. The recipes must be assembled in Coding Canvas and checked on the actual kit. Narration uses Amy Sullivan’s own Voicebox clone. Images are AI-generated mission concepts, not verified LEGO assemblies. Independent teaching draft; not endorsed by the LEGO Group.

Budget your robotics lab

Current kit: LEGO Education Computer Science & AI 6–8 (#45522). US dollars; asking prices and planning allowances are separated below.

Purchase / allowance	USD	Basis
One kit · up to 4 learners	\$529.95	Manufacturer asking price
Six individual kits · 24 learners	\$3,179.70	6 × \$529.95; fewer charging extras than bundle
Classroom bundle · 24 learners	\$3,499.00	6 kits plus charging and replacement extras
Pilot supplies allowance	\$35.00	Planning estimate: paper, tape, ruler and cargo materials
Class supplies allowance	\$120.00	Planning estimate for six teams
Pilot total before tax / shipping	\$564.95	1 kit + \$35 supplies; existing device assumed
Class total before tax / shipping	\$3,619.00	1 bundle + \$120 supplies; 6 existing devices assumed

Prices were checked on the manufacturer's US page. Recheck before ordering. Tax, delivery, devices, school IT support and teacher training are excluded. The supplied course and browser simulator require no paid course subscription. No purchase has been made.

What the classroom bundle adds

The 24-student bundle includes six kits, three additional multi-chargers, eight extra connection cards and a replacement brick pack. Six single kits cost \$319.30 less, but do not provide the same bundle extras.

Choose a purchase plan

Pilot first: one kit, one existing device and four rotating roles. A 24-student class uses six teams of four. For pairs, plan 12 kits: \$6,359.40 for kits alone. An optional spare kit adds \$529.95. All of these are one-time hardware planning figures, not per-student subscription fees.

Older kits are a separate pathway

EV3 and Robot Inventor use different hardware and software. Do not buy them expecting these #45522 recipes to run unchanged. SPIKE direct sales ended June 30, 2026; LEGO lists app support through June 30, 2031. Check used-kit completeness, charging equipment and software before any legacy purchase.

[Selected kit and current US pricing](#)

[SPIKE retirement and support](#)